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MCAUnet: a deep learning framework for automated quantification of body composition in liver cirrhosis patients

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Abstract
Traditional methods for measuring body composition in CT scans rely on labor-intensive manual delineation, which is time-consuming and imprecise. This study proposes a deep learning-driven framework, MCAUnet, for accurate and automated quantification of body composition and comprehensive survival analysis in cirrhotic patients. A total of 11,362 L3-level lumbar CT slices were collected to train and validate the segmentation model. The proposed model incorporates an attention mechanism from the channel perspective, enabling adaptive fusion of critical channel features. Experimental results demonstrate that our approach achieves an average Dice coefficient of 0.952 for visceral fat segmentation, significantly outperforming existing segmentation models. Based on the quantified body composition, sarcopenic visceral obesity (SVO) was defined, and an association model was developed to analyze the relationship between SVO and survival rates in cirrhotic patients. The study revealed that 3-year and 5-year survival rates of SVO patients were significantly lower than those of non-SVO patients. Regression analysis further validated the strong correlation between SVO and mortality in cirrhotic patients. In summary, the MCAUnet framework provides a novel, precise, and automated tool for body composition quantification and survival analysis in cirrhotic patients, offering potential support for clinical decision-making and personalized treatment strategies.

Keywords Convolutional neural network, Body composition quantification, Liver cirrhosis, Sarcopenia visceral obesity, Survival analysis

Introduction

With the rapid development of artificial intelligence technology, computer vision has shown great potential in the field of medical image processing, especially in the diagnosis and management of chronic diseases. Liver cirrhosis, a globally prevalent chronic condition, is responsible for approximately 1.3 million deaths annually due to its associated complications [1]. As the main organ for maintaining metabolic homeostasis, the liver is involved in the absorption and metabolism of glucose, protein and lipids, and plays a vital role in ensuring the body's nutritional supply [2, 3]. Studies have shown that the body composition of patients with cirrhosis is an important factor

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affecting their prognosis and survival rate [4]. However, since patients with cirrhosis often have fluid retention or obesity, traditional nutritional assessment methods, such as the Child-Turcotte-Pugh score [5], are often inaccurate in this population and fail to reliably reflect the patient's true body composition [6, 7]. In recent years, body composition measured by CT scans has become the gold standard and known driver of metabolic syndrome and nutritional status [8]. In addition, more and more people now realize that fat and muscle content are important determinants of patient prognosis [9]. In other words, once patients develop cirrhosis, changes in body composition can become a surrogate indicator of the nutritional status of patients with cirrhosis. However, body composition needs to be clearly defined and quantified for use in clinical care. In this regard, quantification of body composition based on cross-sectional imaging techniques (computed tomography (CT) or magnetic resonance imaging (MRI)) has emerged as an objective and reproducible means of assessing the nutritional status of patients with cirrhosis, as it is not affected by changes in fluid status, bone density, and ascites [10].

So far, there is a large body of evidence that CT-based measurements of muscle and fat tissue distribution have prognostic value for patients with cirrhosis [11–14]. However, the standard method for measuring body composition in CT scans involves tedious manual outlining [15], which is time-consuming and imprecise due to inter- and intra-observer variability [16]. It is also unrealistic to apply it to clinical care. Recent reviews have emphasized the prognostic value of precise body composition quantification, especially for sarcopenia and visceral adiposity, and highlighted the need for scalable, non-invasive imaging-based methods to replace manual segmentation [17]. The rapid development of computer vision technology has inspired scholars to conduct image segmentation from the perspective of deep learning, among which U-Net, as the most widely used network architecture in the field of medical image segmentation, has been continuously optimized and improved by scholars [18, 19]. Attention U-Net [20] enhances important regions via spatial attention but overlooks rich inter-channel dependencies. Residual Attention U-Net [21] improves gradient flow with shortcut connections yet lacks effective multi-scale semantic fusion. Gated Attention U-Net [22] introduces gating mechanisms to refine features but is limited in capturing long-range contextual information. TransUNet (Chen et al. 2021) [23] is the first Transformer-based medical image segmentation framework. Then Swin-Unet (Cao et al. 2022) [24] achieved state-of-the-art performance. The first pure Transformer-based U-type architecture was proposed, which introduced the Swin Transformer to replace the convolutional blocks in U-Net. Although these models improve global feature modeling, their complexity and reliance on large-scale datasets limit practical deployment, which may cause problems such as structural redundancy and high computational cost.

In this work, we proposed the MCAUnet model, a framework for accurate quantification of body composition and survival analysis in patients with cirrhosis. We reconsidered the design of jump connections in Unet to better connect features between the encoder and decoder stages. The framework added an attention mechanism from a channel perspective, which can adaptively fuse sufficient channel features to facilitate complex medical image segmentation. The model was used to predict the body composition of 117 adult patients with cirrhosis who visited Shanxi Bethune Hospital between January 2016 and December 2020. The body composition was quantified based on the segmentation results, and the SVO definition criteria that met the Asian criteria were developed. SVO was defined as sarcopenia (SMI $< 43.24\,\mathrm{cm}^2/\mathrm{m}^2$ for men and SMI $<35.11\,\mathrm{cm}^2/\mathrm{m}^2$ for women) and visceral obesity (VSR ≥ 1.07 for men and VSR ≥ 0.68 for women). Finally, the relationship between SVO and the outcomes of patients with cirrhosis was analyzed. Our main contributions can be summarized as follows:

(1) The MCAUnet model is proposed to achieve efficient and accurate visceral fat segmentation and body composition quantification, and the segmentation performance is significantly better than existing methods.
(2) Sarcopenic visceral obesity (SVO) was defined and confirmed by regression analysis to be an independent prognostic factor for survival in patients with cirrhosis, which has important clinical significance.
(3) A survival analysis model for patients with cirrhosis based on SVO was constructed, providing a scientific basis for patient prognosis assessment and survival rate prediction.

Methods

In this section, we describe our proposed deep learning driven scheme for quantifying body composition and prognosis prediction. Specifically, we first provide a segmentation quantification convolutional network for segmenting visceral fat and quantifying body composition, and finally adopt survival analysis to predict the prognosis of patients with cirrhosis. In the following subsections, we will describe these details in detail.

Measurement of body composition

In this study, we employed deep learning techniques to quantify body composition. As far as we know, Existing transformer-based architectures like TransUNet and Swin-Unet mainly enhance the encoder by introducing global attention or hierarchical representations. However, they generally retain conventional skip connections that directly pass shallow features from encoder to decoder without addressing semantic disparity. These skip connections often bring low-level details (e.g., edges, textures) that lack sufficient semantic context, which can interfere with the decoding process and reduce segmentation precision [25]. In contrast, this paper constructs a MCAUnet framework, as shown in Fig. 1, and designs a Transformer module divided by a channel between an ordinary U-Net encoder and decoder to better integrate encoder features and reduce the semantic gap.

Specifically, we propose a channel transformer (MCA) to replace the skip connection in U-Net, as shown in Fig. 2. Compared with other U-Net variants that aim to improve the skip connection mechanism, MCAUnet focuses on multi-scale feature alignment and adaptive channel-wise attention. By aggregating encoder features at multiple resolutions and modeling inter-channel dependencies, MCA effectively reduces the semantic gap between encoder and decoder representations and achieves more precise feature fusion.

Multi-scale feature embedding: Multi-scale feature fusion is a technique that integrates feature maps of various sizes at the module input stage in order to capture feature information at different spatial levels. This goal can be achieved by using convolution kernels with different field of view sizes or cross-scale pooling operations. Given the output of four skip connection layers $E_i \in R^{\frac{HW}{i^2} \times C_i} (i = 1, 2, 3, 4)$, the features are reshaped into a flattened two-dimensional patch sequence of P, P/2, P/4, and P/8, respectively, so that these patches can be mapped to the same area at four scales of the encoder features. Through this process, we keep the original channel size and then connect the tokens of the four layers.

Multi-head cross attention: Inside the module, the interaction between different channels is introduced through the multi-head attention mechanism. Each attention head can focus on the relationship between different channels, thereby capturing the mutual influence between multi-channel features. Different from the original self-attention, we perform attention along the channel axis instead of the patch axis, and use instance normalization to normalize the similarity matrix of each instance on the similarity map so that the gradient can be propagated smoothly.

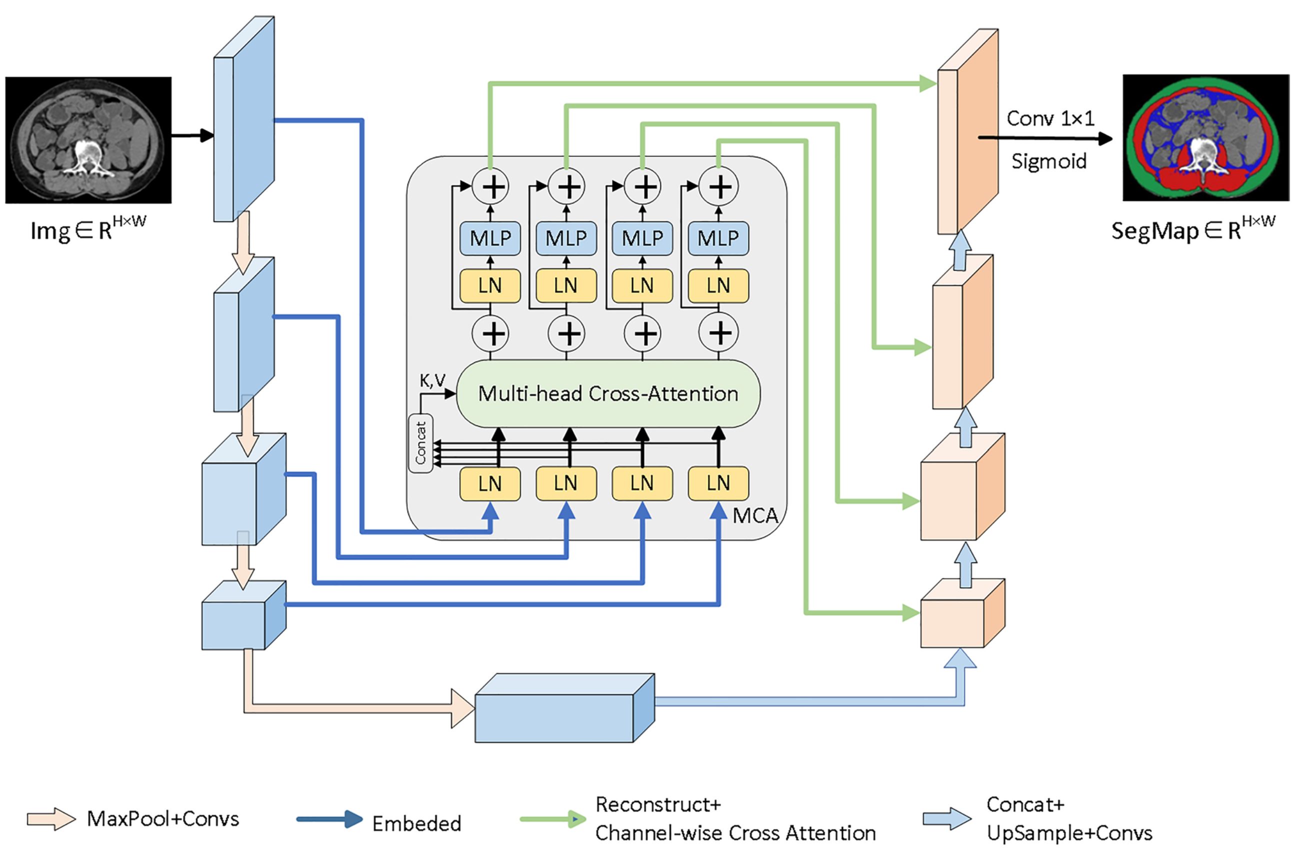

Fig. 1 Model structure of MCAUnet. We replace the original skip connections by a channel converter (MCA)

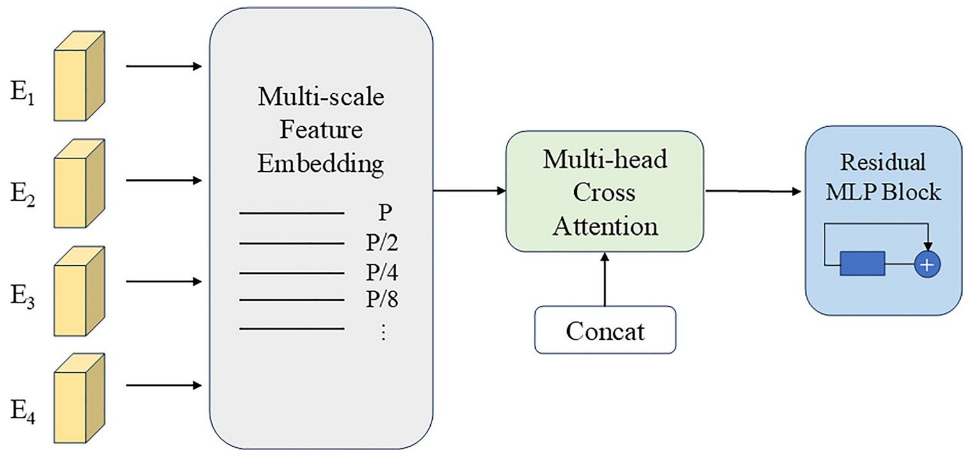

Fig. 2 MCA module. The MCA module integrates multi-scale feature embedding, multi-head channel cross attention, and multi-layer perceptron (MLP) to enhance feature fusion and reduce semantic gaps between encoder and decoder stages

Multilayer Perceptron (MLP): The multilayer perceptron in this network has a residual structure, and information can flow along multiple paths, allowing the network to learn and solve some complex problems more effectively and solve the gradient vanishing problem.

Statistical analysis

Quantitative variables that obey the normal distribution are described by mean ± standard deviation, and quantitative variables that do not obey the normal distribution are described by median ± interquartile range. The categorical variables are described as quantity and proportion. Statistical differences were determined using the *t*-test with Welch correction or the Mann–Whitney *U* test, the Wilcoxon signed-rank test, or the Kruskal-Wallis Row test.

The total cross-sectional area (cm^2) of each body composition compartment was calculated based on the pixel-to-length correspondence in the CT images. All measurements were then normalized by the square of the patient's height (m^2) to obtain the subcutaneous adipose tissue index (SATI, cm^2/m^2), skeletal muscle index (SMI, cm^2/m^2), and visceral adipose tissue index (VATI, cm^2/m^2). VSR was calculated by dividing VATI by SATI. Given the lack of a universally accepted standard for defining low skeletal muscle area and low fat area on CT images, we employed a data-driven approach by defining the cutoff thresholds based on the first quartile values of skeletal muscle and fat area indices within the overall study population. This approach is consistent with previous studies in hepatology and oncology that used internal distribution-based criteria to identify high-risk body composition phenotypes in the absence of international standards [26, 27].

Survival analysis

Survival results were summarized using the Kaplan-Meier method, and each group was compared using a log-rank test. The main results of the study were important predictors of patient death, determined using multiple Cox regression analysis proportional hazard models. Established prognostic factors for clinical death and candidate predictors of $p < 0.10$ in the regression analysis were included in multivariate Cox regression analysis. Bilateral $p < 0.05$ was considered statistically significant. The analysis was performed using SPss25.0 software (IBM, New York, USA) with RStudio version 7.2 (RStudio, New York, USA).

Results

Dataset description

The data used for training and internal validation consisted of 11,362 individual L3-level CT slices provided by the imaging department of Shanxi Bethune Hospital. These were two-dimensional axial images acquired using Siemens SOMATOM Definition Flash and SOMATOM Definition AS CT scanners (Siemens Medical Solutions, Forchheim, Germany). The abdominal scans were performed with the following parameters: tube voltage 120 kV, baseline tube current 120–240 mA, collimation width 0.6 × 128 mm, rotation time 0.5 s per cycle, slice thickness 5 mm, and reconstruction layer thickness 1 mm. Each slice was manually annotated by professional radiologists to generate label maps for subcutaneous fat, skeletal muscle, and visceral fat.

All images were divided into training and internal validation sets in an 8:2 ratio. Both sets included images from a mix of improved and non-improved CT reconstructions, and their equivalence in cross-sectional area measurements was previously verified. Importantly, the patients corresponding to these 11,362 slices were not included in the clinical test cohort, ensuring complete separation between the training/validation and test datasets.

The clinical test dataset was composed of CT scans from an independent cohort of 117 cirrhotic patients at Shanxi Bethune Hospital, used to evaluate model performance and clinical applicability. These patients were retrospectively selected from adults diagnosed with liver cirrhosis between January 2016 and December 2020. Patients with hepatocellular carcinoma or those referred for liver transplantation were excluded. Demographic and clinical data were extracted from medical records, and baseline data were collected within six months of the CT scan. Mortality data were obtained through hospital records and telephone follow-up, with the final follow-up conducted in May 2022. Notably, none of the CT images in the test dataset were used during model training or validation.

Implementation details

We implemented our model with Pytorch on a single NVIDIA A40 GPU with 48 GB memory. To enhance generalization and mitigate device-specific biases, various data augmentation strategies were employed during training, including 90°, 180°, and 270° rotations, horizontal flipping, and contrast normalization. These augmentations simulate the variability in image orientation and acquisition characteristics across different scanners and clinical settings. The single channel and gray image are copied to obtain a three-channel RGB (red, green, and blue) image with a size of $512 \times 512 \times 3$. Each pixel in each image is divided into four categories: subcutaneous fat, skeletal muscle, visceral fat, and background. The model was trained for 100 epochs using the Adam optimizer with an initial learning rate of 1e-3 and a batch size of 16. We also employ the combined cross entropy loss and dice loss function to train our network.To make the results on the small datesets more convincing, we conduct a three times 5-fold cross validation and obtain the mean result and std.

The Dice coefficient and IOU (Intersection Over Union) are used to evaluate the similarity between the predicted results of the model and the true values, respectively. The PPV (Positive Predictive Value) and specificity are used to evaluate the ability of the model to predict the correct results.

Mcaunet model segmentation results

The segmentation results of the model are shown in Fig. 3. The artificially delineated Ground truth provides the basic facts for training the deep learning model, and finally the model can automatically segment and measure subcutaneous fat (green part), skeletal muscle (red part), and visceral fat (blue part).

To evaluate the performance of the proposed MCAUnet model, we compared it with both traditional segmentation models, including U-Net, PSPNet, and DeepLabV3+, as well as several recently developed state-of-the-art architectures, such as Attention U-Net,

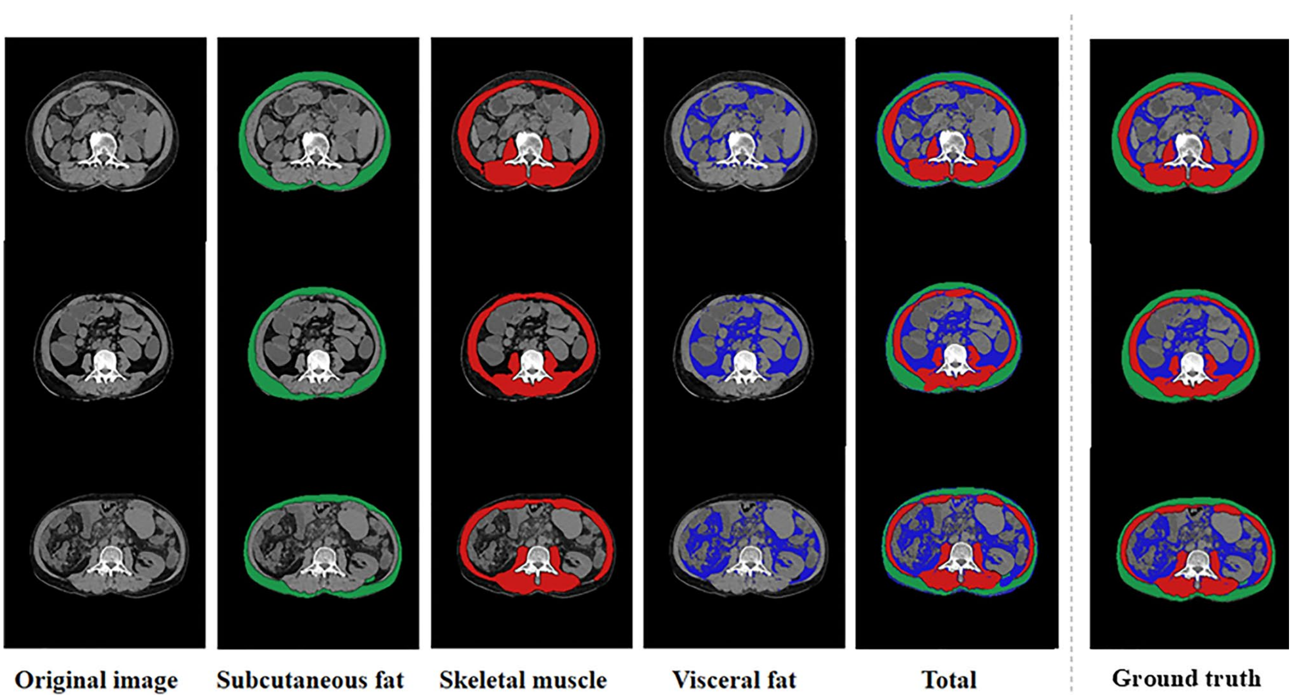

Fig. 3 Segmentation results of the MCAUnet model, showing the automated delineation of subcutaneous fat (green), skeletal muscle (red), and visceral fat (blue)

Table 1 Quantitative results of models on test datasets

Model	Dice	IOU	PPV	Specificity
U-Net	0.928	0.867	0.886	0.878
PspNet	0.746	0.678	0.762	0.770
DeeplabV3+	0.906	0.830	0.891	0.887
Attention U-Net	0.932	0.875	0.900	0.882
TransUNet	0.936	0.885	0.930	0.920
Swin-Unet	0.948	0.898	0.942	0.933
MCAUnet	**0.952**	**0.908**	**0.952**	**0.950**

Table 2 Comparison of computational complexity between different models

Model	Parameters (M)	Inference time (ms/sample)	GPU memory (GB)
U-Net	31	22	4.5
PSPNet	65	28	6.0
DeepLabV3+	55	30	5.8
Attention U-Net	35	24	5.0
TransUNet	97	45	8.5
Swin-Unet	92	43	8.0
MCAUnet (Ours)	**38**	**25**	**5.2**

Table 3 Results of ablation experiments

Model	Dice	IOU	PPV	specificity
MCAUnet	**0.952**	**0.908**	**0.952**	**0.950**
Without multi-scale embed	0.937	0.882	0.935	0.941
Without channel attention	0.929	0.874	0.927	0.938
Without MLP block	0.943	0.892	0.942	0.943

TransUNet, and Swin-Unet. All models were evaluated on a clinical validation set comprising 185 samples. As summarized in Table 1, MCAUnet achieved the best overall performance across all evaluation metrics. Specifically, the Dice coefficient reached 0.952, and the IoU reached 0.908. The PPV and specificity also attained optimal values of 0.952 and 0.950, respectively. These results demonstrate that MCAUnet not only addresses the limitations of previous models but also achieves more accurate and robust segmentation performance, particularly in complex clinical imaging scenarios.

To further evaluate the practical feasibility of deploying different models in clinical settings, we also compared their computational complexity in terms of model size, inference time, and GPU memory usage. As summarized in Table 2, MCAUnet exhibits a balanced trade-off between segmentation accuracy and computational efficiency, making it suitable for clinical applications requiring real-time or resource-constrained conditions.

Ablation study

To investigate the contribution of each key component in the proposed MCA module, we conducted an ablation study. Three variants were constructed by selectively removing the multi-scale embedding, channel attention, and MLP block from the MCA module. All variants were trained and evaluated under the same protocol used for the full model. As shown in Table 3, the removal of multi-scale feature embedding or multi-head cross attention leads to a substantial drop in segmentation performance, suggesting that both are critical for contextual understanding and semantic alignment. The MLP block also plays a notable role by stabilizing feature transformation. These findings support the architectural design choices of the MCA module.

Results of human body composition

To evaluate the clinical relevance and practicability of the deep learning model, we quantified the segmentation results in the previous section and calculated the distribution of SATI, SMI, and VATI in different genders as shown in Fig. 4. Since the values of L3SATI, L3SMI, and L3VATI are all normal distribution variables, the first quartile is used as the cut-off value of the low subcutaneous fat index, low skeletal muscle index, and low visceral fat index in this study. The definitions of the low subcutaneous fat index, low skeletal muscle index, and low visceral fat index in different genders are as follows. L3SATI < $20.32\,\text{cm}^2/\text{m}^2$ and L3SATI < $34.09\,\text{cm}^2/\text{m}^2$ in both males and females had low subcutaneous fat levels. L3SMI < $43.24\,\text{cm}^2/\text{m}^2$ and L3SMI < $35.11\,\text{cm}^2/\text{m}^2$ in both males and females had low skeletal muscle levels. L3VATI < $25.74\,\text{cm}^2/\text{m}^2$ and L3VATI < $24.42\,\text{cm}^2/\text{m}^2$ L3SMI < $43.24\,\text{cm}^2/\text{m}^2$ and L3SMI < $35.11\,\text{cm}^2/\text{m}^2$ in both males and females had low visceral fat levels.

The clinicopathological features of the study population

Studies have shown that based on CT muscle and adipose tissue measurements combined to define SVO can better determine the morphological characteristics of obesity associated with the outcome of the disease. We use the ratio of visceral fat to subcutaneous fat (VSR) to assess visceral obesity. To determine the optimal threshold for VSR to define visceral obesity, we first assessed the relationship between mortality and VSR based on differences in visceral and subcutaneous fat observed in our cohort. Based on the VSR quantified previously, we used the VSR cut-off point of $male \geq 1.07$ and $female \geq 0.68$ to define visceral obesity. SVO was defined as a combination of sarcopenia (male SMI < $43.24\,\text{cm}^2/\text{m}^2$, female SMI < $35.11\,\text{cm}^2/\text{m}^2$) and visceral obesity (male $VSR \geq 1.07$, female $VSR \geq 0.68$).

We divided 117 patients in the dataset into SVO patients and Non-SVO patients. As shown in Table 4, there were 22 (18.8%) patients with SVO and 95 (81.2%) patients with Non-SVO. Baseline clinical characteristics of patients with and without SVO are also shown in Table 4. There was no difference in sex ratio and age between

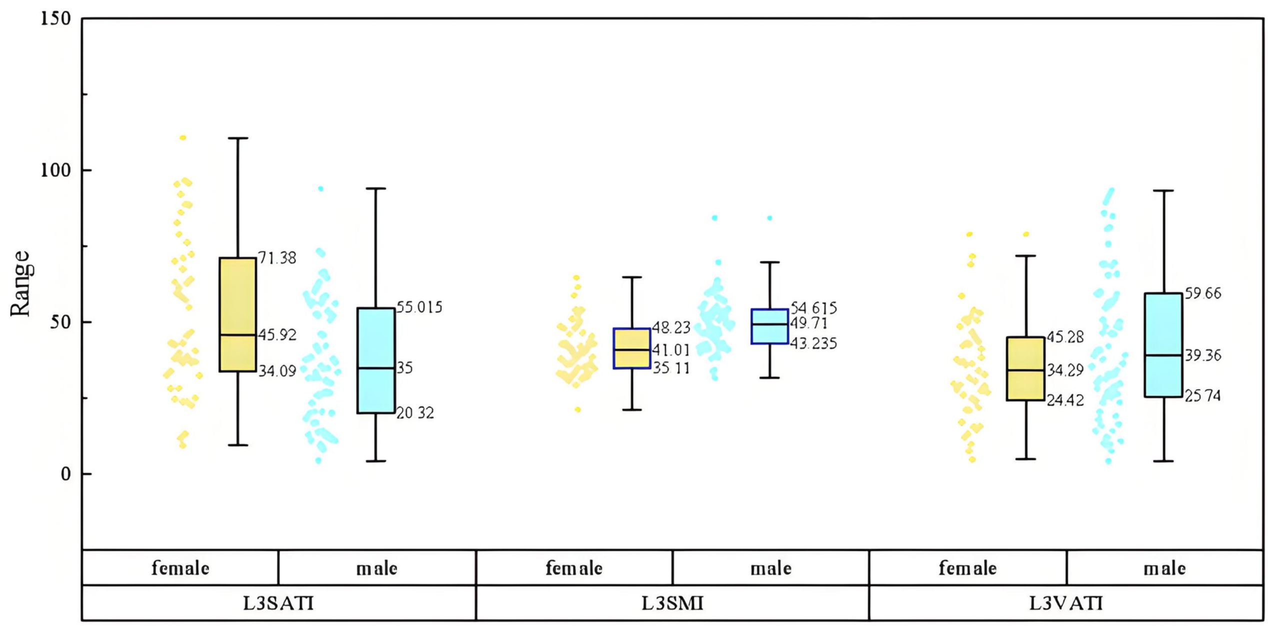

Fig. 4 Distribution and demarcation of L3SATI, L3SMI, and L3VATI in different gender

the two groups. In the SVO group, BMI was significantly lower ($p = 0.001$), and body weight was significantly lighter ($p = 0.008$). At the same time, the concentrations of certain metabolic rate markers were lower, such as lower creatinine (Cre) ($p = 0.047$) and lower prothrombin time (PT) ($p = 0.044$). More importantly, the MELD score was higher in the SVO group ($p = 0.003$), which represented a higher severity of cirrhosis.

Survival analysis of SVO patients

Recognition of body composition as an important determinant of mortality in patients with cirrhosis [28], We used deep learning models to extract body composition features from clinical CT scans to predict mortality. In Cox proportional hazard regression analysis, we found that many components of body composition were significantly associated with mortality. As shown in Table 5, combining HR and p-values, Cox regression analysis shows that BMI (HR 1.06, 95% CI 1.02–1.1, $p = 0.03$), prothrombin time (PT-S, HR 1.08, 95% CI 1.11–1.52, $p = 0.016$), international normalized ratio (INR, HR 2.38, 95% CI 1.2–4.8, $p = 0.014$), prothrombin activity (PTA, HR 0.96, 95% CI 0.93–0.98, $p = 0.022$), model of end-stage liver disease (MELD, HR 1.04, 95% CI 1.01–1.1, $p = 0.043$), alkaline phosphatase (ALP, HR 0.96, 95% CI 0.94–0.99, $p = 0.028$), sarcopenic visceral obesity (SVO, HR 1.14, 95% CI 1.02–2.97, $p = 0.038$) was significantly associated with overall survival rate (OS). In multivariate Cox regression, SVO was still independently associated with OS. As shown in Table 4, sarcopenic visceral obesity (SVO, HR 1.53, 95% CI 1.31–4.38, $p = 0.046$) and alkaline phosphatase (ALP, HR 0.69, 95% CI 0.45–0.97, $p = 0.023$) were independent prognostic factors for OS in patients with cirrhosis.

Next, we performed a survival analysis of patients with SVO. During the 5-year follow-up, no patients were lost to follow-up and 28 patients died. As shown in Fig. 5, the 3-year cumulative survival rate of cirrhotic patients with or without SVO was 32% and 50%, respectively and the 5-year cumulative survival rate of cirrhotic patients with or without sarcopenic obesity was 5% and 32%, respectively. This indicates that the cumulative incidence of death in cirrhotic patients with sarcopenic obesity was significantly higher. The presence of SVO was significantly associated with the cumulative incidence of mortality in patients with cirrhosis ($p = 0.035$).

Comparison of different standard methods

We evaluated the improvement in predicting survival rate of patients when using VSR to define visceral obesity. Compared with other criteria used in previous studies, we used the MELD (model for end-stage liver disease) scoring system. MELD $\leq$ 14 is defined as low-risk, and MELD > 14 is defined as intermediate-risk or high-risk patients. It can be seen that there is no statistically significant relationship with the survival rate of patients with cirrhosis as shown in Fig. 6.

In addition, if the patients with SVO were defined by the sum of fat area and sarcopenia, which we redefined as SFO (total fat area index $(TFAI) \geq 46.06\,\mathrm{cm}^2/\mathrm{m}^2$ in males, $TFAI \geq 58.51\,\mathrm{cm}^2/\mathrm{m}^2$ in females, $\mathrm{SMI} < 43.24\,\mathrm{cm}^2/\mathrm{m}^2$ in males, $\mathrm{SMI} < 35.11\,\mathrm{cm}^2/\mathrm{m}^2$ in females), it can be seen that there is no statistically significant relationship between the survival rate of patients

Table 4 Clinical demographic and laboratory characteristics of sarcopenic visceral obesity

Variable	Overall *n*=117	Non-sarcopenic visceral obesit *n*=95	Sarcopenic visceral obesity *n*=22	*P*-value
sex = 1%)	68 (58.1)	56 (58.9)	12 (54.5)	0.891
Age	55.93 (11.05)	55.71 (10.91)	56.91 (11.86)	0.647
BMI	24.71 (3.59)	25.25 (3.52)	22.37 (2.92)	0.001
Weight	65.50 [59.10, 77.50]	67.50 [60.10, 78.30]	62.00 [49.50, 68.20]	0.008
Height	164.80 (8.75)	164.97 (8.46)	164.09 (10.10)	0.674
SMI	46.82 [40.26, 52.94]	48.96 [42.63, 53.75]	34.27 [32.45, 41.19]	< 0.001
VSR	0.93 [0.70, 1.31]	0.87 [0.61, 1.25]	1.26 [1.01, 1.63]	0.001
ALT	56.88 (68.53)	60.71 (73.41)	40.30 (38.24)	0.209
AST	58.30 [35.70, 96.70]	58.30 [38.70, 83.55]	57.00 [29.68, 108.95]	0.53
ALP	117.40 [83.50, 146.80]	109.20[84.35, 144.50]	130.90[85.10,192.73]	0.261
r-GT	78.80 [39.90, 163.90]	68.20 [37.50, 146.90]	123.65[48.45,223.52]	0.149
TBIL	32.60 [21.50, 69.50]	31.40 [21.85, 74.80]	33.05 [20.62, 60.25]	0.563
DBIL	10.90 [5.80, 25.70]	10.40 [5.80, 27.35]	12.15 [4.40, 16.75]	0.688
IBIL	20.40 [15.60, 43.90]	21.50 [16.00, 48.55]	18.65 [12.47, 36.95]	0.25
Alb	30.58 (5.78)	30.72 (5.93)	29.99 (5.18)	0.597
Cre	67.60 [57.30, 77.30]	69.20 [58.70, 77.30]	59.50 [49.65, 70.65]	0.047
BUN	4.80 [3.80, 6.10]	4.90 [3.90, 6.15]	4.05 [3.30, 5.72]	0.238
PT-S	14.90 [13.00, 16.90]	15.00 [13.30, 16.90]	13.70 [12.33, 15.38]	0.044
INR	1.37 [1.20, 1.55]	1.38 [1.23, 1.56]	1.27 [1.14, 1.42]	0.043
PTA	62.00 [51.00, 76.00]	60.00 [50.00, 72.50]	73.50 [59.00, 79.75]	0.068
WBC	3.90 [2.70, 5.40]	3.90 [2.70, 5.10]	4.20 [2.65, 5.57]	0.558
NAP	2.25 [1.48, 3.27]	2.21 [1.47, 3.18]	2.73 [1.64, 3.58]	0.345
TLC	1.07 (0.70)	1.07 (0.71)	1.04 (0.64)	0.864
RBC	3.57 (0.78)	3.61 (0.78)	3.39 (0.77)	0.228
PLT	76.00 [54.00, 114.00]	69.00 [54.50, 112.50]	82.00 [54.00, 145.75]	0.307
CPG (%)				0.355
A	26 (22.2)	22 (23.2)	4 (18.2)	
B	48 (41.0)	36 (37.9)	12 (54.5)	
C	43 (36.8)	37 (38.9)	6 (27.3)	
CPS	9.00 [7.00, 11.00]	9.00 [7.00, 11.00]	8.00 [7.00, 9.75]	0.487
MELD	7.82 [5.12, 11.60]	4.15 [−0.69, 8.64]	8.02 [6.17, 11.99]	0.003

SMI: skeletal muscle index; VSR: Ratio of visceral to subcutaneous adipose tissue; ALT: glutamic pyruvic transaminase; AST: glutamic-oxalacetic transaminase; ALP: alkaline phosphatase; r-GT: Bile duct related enzymes; TBIL:total bilirubin; DBIL: conjugated bilirubin; IBIL: indirect bilirubin; Alb: albumin; Cre: creatinine; BUN: blood urea nitrogen; PT-S: prothrombin time; INR: international normalized ratios; PTA: prothrombin activity; WBC: white bloodcell count; NAP: neutrophil counts; TLC: lymphocyte numbers; RBC: red-cell count; PLT: platelet count; CPG: Child-Pugh grading standard; CPS: Child-Pugh score; MELD: a model for end-stage liver disease

with cirrhosis as shown in Fig. 7. It shows that the use of SVO to distinguish patients to predict the overall survival of patients with cirrhosis has high reliability.

Discussion

In this study, we introduce a deep learning-based framework, MCAUnet, for automated body composition quantification and survival analysis in cirrhotic patients The model reconsiders the design of jump connections in Unet to better connect features between encoder and decoder stages, and incorporates an attention mechanism from a channel perspective that can adaptively fuse multi-channel features favoring complex medical image segmentation. By analyzing lumbar L3-level CT images of 117 adult cirrhotic patients who were admitted to Shanxi Baiqun Hospital between January 2016 and December 2020, we achieved precise quantification of the body composition of cirrhotic patients and defined the muscle-reduced visceral obesity (SVO) criterion based on Asian populations, which further validated the close relationship between SVO and adverse outcomes of cirrhotic patients The results of Cox regression analysis showed that SVO was a significant independent risk factor affecting the survival of cirrhotic patients. SVO quantification may inform preoperative risk stratification, trigger early nutritional or exercise-based interventions, or help prioritize patients for more intensive monitoring or liver transplant evaluation. As such, automated assessment of SVO has the potential to shift clinical management from reactive to proactive strategies in cirrhotic care.

In previous studies, cirrhotic patients were mostly associated with complications such as fluid retention, which led to a large deviation between body composition results based on measurements by conventional means

Table 5 Cox regression analysis of overall survival rate

	Univariable analysis			Multivariable analysis		
	HR	95% CI	*P*-value	HR	95% CI	*P*-value
sex	1.31	0.85–2	0.219			
Age	0.99	0.97–1.13	0.370			
BMI	1.06	1.02–1.1	0.030			
SMI	1.01	0.97–1.06	0.278			
VSR	1.10	0.71–1.7	0.667			
PT-S	1.18	1.11–1.52	0.016			
AST	0.99	0.79–1.32	0.235			
INR	2.38	1.2–4.8	0.014			
Alb	0.98	0.55–1.32	0.386			
BUN	1.10	0.92–1.5	0.090			
PTA	0.96	0.93–0.98	0.022			
WBC	1.06	0.98–1.2	0.118			
NAP	1.09	0.95–1.2	0.070			
TLC	1.06	0.72–1.6	0.765			
RBC	0.79	0.59–1.1	0.107			
MELD	1.04	1.01–1.1	0.043			
PLT	0.79	0.54–1.34	0.211			
CPS	1.09	0.94–1.25	0.089			
ALP	0.96	0.94–0.99	0.028	0.69	0.45–0.97	0.023
SVO	1.14	1.02–2.97	0.038	1.53	1.31–4.38	0.046

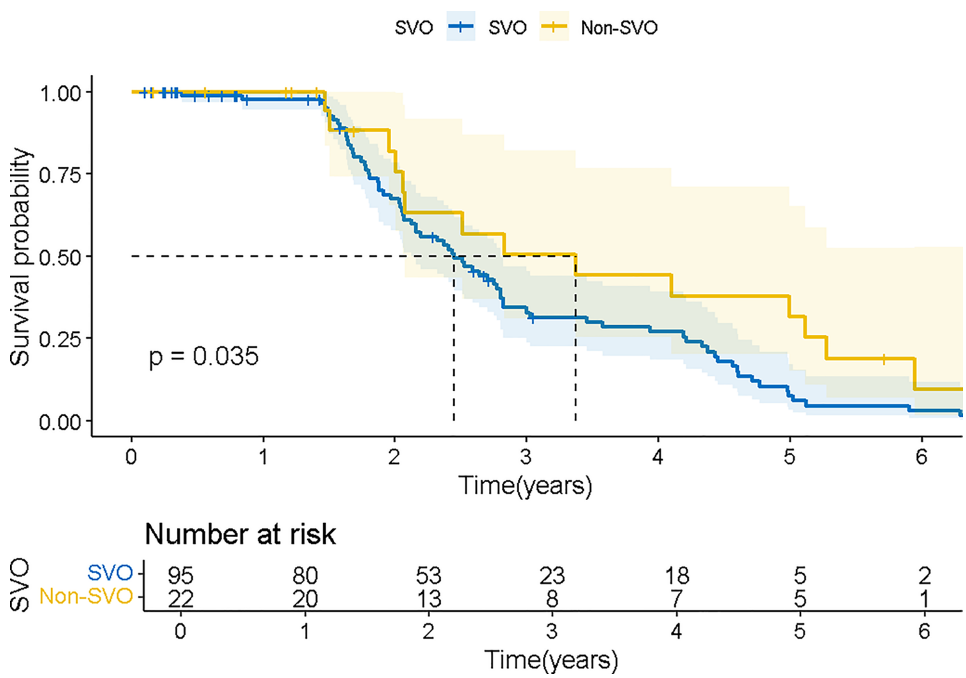

Fig. 5 Survival analysis of patients with liver cirrhosis differentiated by muscular reduction visceral obesity

and the true results. In order to obtain more accurate body composition measurements, cross-sectional imaging of the patient's third lumbar spine CT is becoming a key tool for body composition measurements, especially muscle and fat content [29, 30]. However, standard methods for measuring body composition include visual inspection and time-consuming manual outlining, which are not suitable for clinical implementation. In this study, we propose a new deep learning model, MCAUnet, that enables objective measurement of body composition in CT scans and, more importantly, this process can be fully automated using deep learning methods. We demonstrate that our convolutional neural network is able to

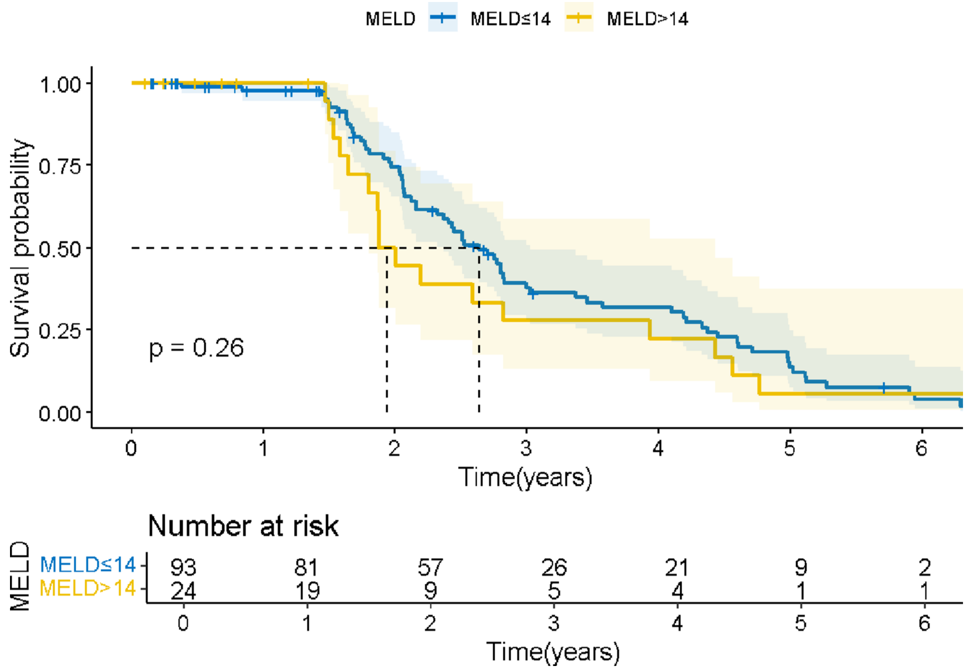

Fig. 6 Survival analysis of patients with liver cirrhosis by MELD score

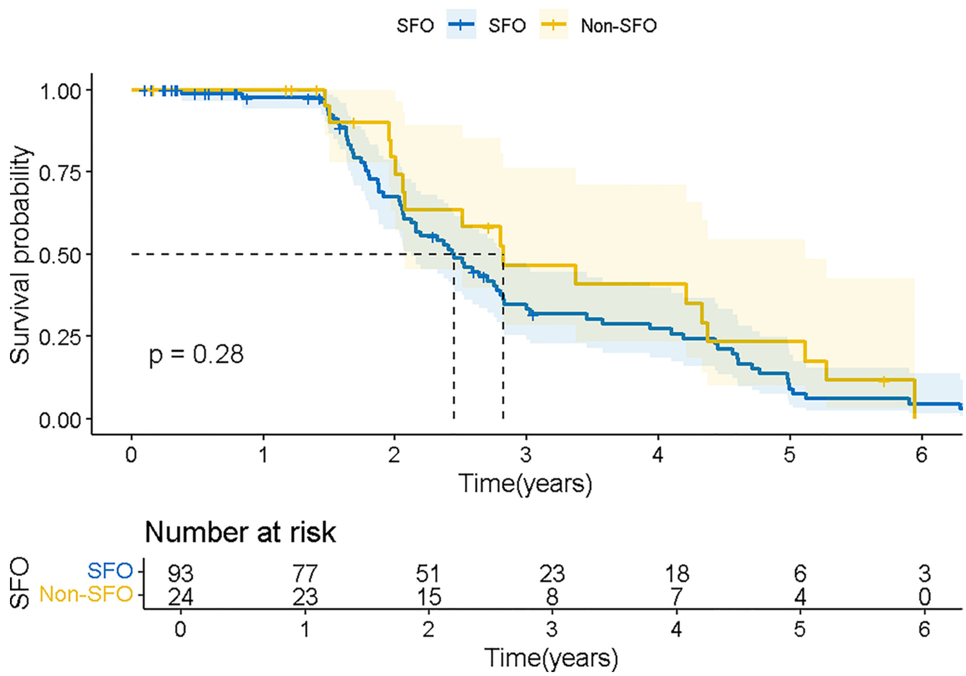

Fig. 7 Survival analysis of patients with liver cirrhosis based on total fat content. SFO were defined by the sum of fat area (total fat area index ($TFAI$) $\geq$ 46.06 cm^2/m^2 in males, $TFAI \geq 58.51$ cm^2/m^2 in females, SMI < 43.24 cm^2/m^2 in males, SMI < 35.11 cm^2/m^2 in females)

accurately identify key regions of the measurement and shows high correlation with manual measurements.

By automating the measurements, we quantified the amount of subcutaneous fat, skeletal muscle, and visceral fat in patients with cirrhosis and defined critical values. In this study, we observed that nearly half of the cirrhotic patients had sarcopenia with or without visceral obesity, as determined by CT-quantified body composition, and 18.8% of these patients were classified as SVO due to the presence of both sarcopenia and visceral obesity.We further observed that, compared to patients with non-sarcopenic visceral obesity, the presence of SVO was associated with a lower survival rate independently associated, resulting in 3- and 5-year survival rates of 32% and 5%, respectively.

Although the exact pathogenesis of the sex difference between sarcopenia and survival is unclear, our findings raise the possibility that SVO, rather than sarcopenia alone, may be a more appropriate indicator to use in clinical practice to identify patients with decompensated cirrhosis who are at increased risk of adverse effects following illness [31]. And we used a deep learning approach to quantify body composition, a new metric that combines muscle and fat quantification to advance our understanding of practical risk assessment in this population.

Furthermore, although the MCAUnet model was trained on single-modality CT images, this study integrated multi-modal analysis by combining the image-derived body composition indices with clinical follow-up data to perform survival analysis. This fusion of imaging and clinical information demonstrates the potential of MCAUnet not only as a segmentation tool, but also as a clinically meaningful risk assessment model, which could be extended in the future to incorporate laboratory, genomic, or functional data.

Our study has several limitations. First, the study population was derived solely from Shanxi Baiqiu'en Hospital, resulting in a relatively small and potentially unrepresentative sample of the broader cirrhotic patient population. However, the observed prevalence of sarcopenia, visceral obesity, and SVO in our cohort was consistent with prior studies, and the median SMI ($43.23\,cm^2/m^2$) closely aligned with previously reported values in Chinese populations ($44.77\,cm^2/m^2$) [32], lending support to the reliability of our findings. Additionally, the generalizability of the MCAUnet model remains to be validated, as it was trained on data from a single-center, single-vendor CT scanner, which may limit its performance across different imaging protocols, scanner types, and institutions. Such device- and site-specific variations can lead to feature instability and performance degradation, as highlighted in recent studies on batch effects in radiomics models [33]. To mitigate scanner- or institution-specific bias, standard data augmentation techniques–such as random affine transformations and intensity jittering–were applied during training. Moreover, we employed three repetitions of 5-fold cross-validation to improve reliability and reduce overfitting on this limited dataset. Still, we acknowledge the lack of external validation. Future studies will prioritize evaluation on multi-center, multi-vendor cohorts and publicly available datasets to enhance model reproducibility and robustness.

Second, although the MCAUnet model shows promise, broader clinical implementation of AI tools in hepatology still faces practical challenges–including dataset heterogeneity, regulatory and ethical concerns, and the "bench-to-bedside" translation gap [34]. practical barriers such as image standardization, integration into hospital PACS or EMR systems, model interpretability, and clinician acceptance must be addressed to facilitate real-world adoption. Moreover, due to the limited number of deaths ($n = 28$) in our cohort, the survival analysis may be subject to potential overfitting in Cox regression. To mitigate this, we applied standard strategies such as variable selection based on univariate screening and clinical relevance, and ensured an event-per-variable (EPV) ratio close to or above the recommended threshold of 10. Nevertheless, internal validation techniques such as bootstrapping or penalized regression (e.g., LASSO) were not implemented and should be considered in future larger-scale studies.

Conclusions

In this study, we developed MCAUnet, a deep learning framework for automated body composition quantification and survival analysis in cirrhotic patients. The model achieves superior performance in complex medical image segmentation by optimizing U-Net's skip connections and incorporating a channel attention mechanism. Based on the model segmentation results, our defined muscle-reduced visceral obesity (SVO) is shown to be an independent predictor of survival in cirrhotic patients, which can provide a scientific basis for predicting patient survival and a strong technical support for personalized diagnosis and treatment.

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Author contributions
W.J was involved in the development of the methodology, the preformance of experiment and wrote the initial draft of the manuscript. X.S participated in data collection and processing and provided significant input in revising the manuscript. Z.J and W.X contributed to the design of the experiments and the interpretation of the results. Z.C and Z.W were responsible for supervision and revising of the document. All authors have given their final approval and agreed to be fully responsible for all aspects of the study, ensuring the integrity and accuracy of the research.

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Data availability
The full data are available upon reasonable request by contacting the corresponding author.

Declarations

Ethics approval and consent to participate
This study was approved by the Shanxi Bethune Hospital (Shanxi Academy of Medical Sciences) Medical Ethics Committee (approval number: YXLL-2022-094). The authors confirm that all experiments were conducted in accordance with relevant guidelines and regulations and that informed consent was obtained from all participants.

Consent for publication
Not applicable.

Competing interest
The authors declare no competing interests.

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